Sample Paper 6 — Class 4 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 35 questions / 40 marks / 60 minutes / no negative marking as Papers 1–5, but noticeably harder throughout — closer to SOF IMO **Level 2**. Attempt this only after scoring comfortably on Sample Paper 5.

Section 1 — Logical Reasoning (Q1–Q10, 1 mark each)

Q1. Find the odd one out: A) Whale B) Dolphin C) Bat D) Shark

Q2. Thermometer is to Temperature as Speedometer is to ___. A) Distance B) Speed C) Time D) Direction

Q3. If each letter of a word is moved 2 steps forward in the alphabet (wrapping Z→B, Y→A), MOUSE becomes OQWUG. Using the same rule, what does KEYBOARD become? A) MGADQCTF B) MGADQCTE C) MFADQCTF D) NGADQCTF

Q4. This series follows two patterns at once — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 1, 4, 3, 8, 5, 12, 7, ___ A) 14 B) 15 C) 16 D) 9

Q5. Five friends sit in a row. Bala sits exactly in the middle. Aman sits at the left end. Chetan sits between Aman and Bala. Esha sits between Bala and Dinesh. In what order are they seated, left to right? A) Aman, Chetan, Bala, Esha, Dinesh B) Aman, Bala, Chetan, Dinesh, Esha C) Chetan, Aman, Bala, Dinesh, Esha D) Aman, Chetan, Bala, Dinesh, Esha

Q6. Priya walks 8 m North, then 6 m East, then 8 m South, then 6 m West. Where is she now relative to her starting point? A) 8 m North B) 6 m East C) Back at the starting point D) 14 m away

Q7. If the 3rd day of a month is a Monday, what day of the week is the 24th of the same month? A) Sunday B) Monday C) Tuesday D) Saturday

Q8. Find the odd one out: A) 8 B) 27 C) 64 D) 100 (Hint: three of these follow the same multiplication pattern — a number multiplied by itself twice.)

Q9. Find the next number: 1, 2, 6, 24, 120, ___ A) 240 B) 480 C) 600 D) 720

Q10. In a row, Neel is 6th from the left and 9th from the right. If two more children join the row at the right end, what will be Neel's position from the right now? A) 9th B) 10th C) 11th D) 12th

Section 2 — Mathematical Reasoning (Q11–Q20, 1 mark each)

Q11. 34,567 + 28,945 = ? A) 63512 B) 63412 C) 63612 D) 62512

Q12. 50,000 − 23,678 = ? A) 26322 B) 26332 C) 26422 D) 27322

Q13. 1,234 × 6 = ? A) 7404 B) 7304 C) 7414 D) 7044

Q14. 9,576 ÷ 12 = ? A) 798 B) 788 C) 808 D) 796

Q15. Find the HCF of 36 and 48. A) 6 B) 8 C) 12 D) 24

Q16. Find the LCM of 8 and 12. A) 16 B) 20 C) 24 D) 96

Q17. Which of these fractions is the greatest: 5/6, 7/9, 3/4, 4/5? A) 5/6 B) 7/9 C) 3/4 D) 4/5

Q18. 2 km 450 m + 1 km 800 m = ? A) 3 km 250 m B) 4 km 250 m C) 4 km 50 m D) 3 km 950 m

Q19. A rectangular field is 24 m long and 18 m wide. Find its area. A) 408 sq m B) 420 sq m C) 432 sq m D) 442 sq m

Q20. A regular pentagon has a perimeter of 45 cm. Find the length of each side. A) 8 cm B) 9 cm C) 10 cm D) 7 cm

Section 3 — Everyday Mathematics (Q21–Q30, 1 mark each)

Q21. A school collected ₹45,600 from 3 classes equally. How much did each class contribute? A) ₹15,000 B) ₹15,100 C) ₹15,200 D) ₹15,300

Q22. A factory makes 1,440 bottles in 8 hours, at a steady rate. How many bottles does it make in 3 hours? A) 520 B) 540 C) 560 D) 500

Q23. A tank holds 2,400 L. It is filled by a pipe at 150 L per minute. How long will it take to fill the tank completely? A) 14 min B) 15 min C) 16 min D) 18 min

Q24. A bus leaves city A at 7:20 a.m., stops for 25 minutes on the way, and reaches city B at 12:05 p.m. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 10 min B) 4 h 20 min C) 4 h 30 min D) 4 h 45 min

Q25. 2 L 500 mL of one liquid is mixed with 1 L 750 mL of another, then poured equally into 5 bottles. How much does each bottle get? A) 800 mL B) 850 mL C) 900 mL D) 750 mL

Q26. Rohit buys 6 notebooks at ₹28 each and gets a discount of ₹18 on the total bill. How much does he pay? A) ₹140 B) ₹150 C) ₹160 D) ₹168

Q27. A cistern with 840 L of water is used for irrigation at the rate of 1/7 of its original capacity every day. How much water remains after 2 days? A) 540 L B) 580 L C) 600 L D) 620 L

Q28. A delivery van carries 3 boxes weighing 18 kg 500 g, 22 kg 750 g, and 15 kg 250 g. Find the total weight carried. A) 55 kg 500 g B) 56 kg 500 g C) 56 kg D) 57 kg 500 g

Q29. 1,080 chairs are arranged equally into rows of 24 chairs each. How many rows will there be? A) 40 B) 42 C) 45 D) 48

Q30. A trader bought 50 kg of sugar for ₹2,000 and sold all of it at ₹45 per kg. Find his total profit. A) ₹200 B) ₹250 C) ₹300 D) ₹225

Section 4 — Achievers Section (Q31–Q35, 2 marks each)

Q31. A number is first multiplied by 5, then 12 is subtracted from the result, giving 63. Find the original number. A) 13 B) 14 C) 15 D) 16

Q32. The sum of three numbers is 150. The second number is twice the first, and the third is 10 more than the second. Find the first number. A) 24 B) 26 C) 28 D) 30

Q33. Anjali had a certain amount of money. She spent 1/3 of it on books, then 1/4 of what remained on stationery, and was left with ₹900. How much money did she have at first? A) ₹1,500 B) ₹1,600 C) ₹1,800 D) ₹2,000

Q34. In a school, the number of boys is 5 more than twice the number of girls. There are 320 students in total, and 105 of them are girls. Find the number of boys. A) 205 B) 210 C) 215 D) 220

Q35. A sequence starts with 5 and 7, and each later term is the sum of the two terms before it: 5, 7, 12, 19, 31, ___. Find the next term. A) 42 B) 45 C) 48 D) 50

Answer key: `sample-paper-06-answer-key.md`
